

Nexivarin 10 mg Tablets — eCTD Submission
Module 4 — Nonclinical Study Reports

Submission Type	NDA — 505(b)(1)	Sequence Number	0000
Submission Date	2021-03-15	Section	4.2.3.3
Prepared by	PharmaCorp Inc.	Confidentiality	Proprietary and Confidential

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Section	Title	Description
4.2.3.3	Genotoxicity Studies	ICH S2(R1) standard genotoxicity battery
Study NXV-GEN-001	Bacterial Reverse Mutation (Ames Test)	Negative in S. typhimurium and E. coli ± S9
Study NXV-GEN-002	In Vitro Chromosomal Aberration (CHO)	Negative up to cytotoxic concentrations
Study NXV-GEN-003	In Vivo Micronucleus Test (Mouse Bone Marrow)	Negative at doses up to 2000 mg/kg

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NDA 123456

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Section 4.2.3.3
Genotoxicity

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4.2.3.3 Genotoxicity Studies — Nexivarin Hydrochloride

A standard battery of genotoxicity tests was conducted per ICH S2(R1):

Study No.	Test System	Doses/Concentrations	Result	GLP
NXV-GEN-001	Ames test (Salmonella typhimurium TA98, TA100, TA1535, TA1537; E. coli WP2 uvrA) ± S9	1.6 – 5000 µg/plate	Negative (all strains, ± S9 activation)	Yes
NXV-GEN-002	In vitro chromosome aberration assay (CHO cells) ± S9, 24h treatment	50 – 2000 µg/mL	Negative (no statistical increase in aberrant cells at any concentration)	Yes
NXV-GEN-003	In vivo mouse micronucleus test (bone marrow, CD-1 mouse)	250, 500, 1000 mg/kg (single oral dose)	Negative (MN frequency not significantly different from vehicle control at any dose; positive control valid)	Yes

Overall Genotoxicity Assessment: Nexivarin hydrochloride is non-genotoxic. The compound did not induce mutations, chromosomal aberrations, or micronuclei in the standard ICH S2(R1) battery. These results, combined with the absence of structural alerts for genotoxicity in the chemical structure, confirm that nexivarin does not pose a genotoxic risk.